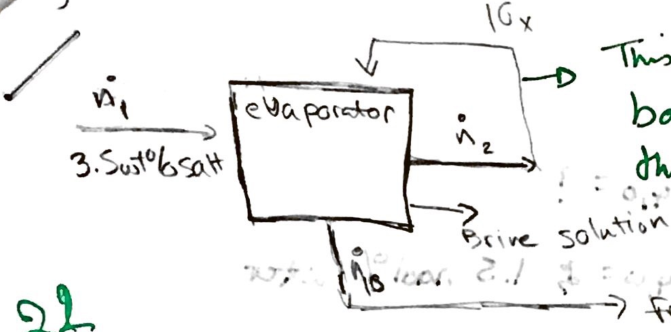
3.5 wt% salt

a)



This exits and comes right back in. doesn't change any thing in the balances only makes it more difficult.

22/25

b) Basis: 1 Kg/s of water is used

$\dot{n}_1 = 10 n_2 + \dot{n}_{3,10}$ $\qquad$ $\dot{n}_{1,w_0} = 10 n_2 + \dot{n}_{3,w_{10}}$ $\qquad$ $\dot{n}_{1,s_1} = \dot{n}_{3,s_{10}}$

$1 = 10 n_2 + \dot{n}_3$ $\qquad$ $.965 = 10 n_2 + .95 \dot{n}_3$ $\qquad$ $.035 = .05 \dot{n}_3$

for 4th Evaporator

$\dot{n}_{1,s_1} = \dot{n}_{3,s_4} \longrightarrow .035 = \dot{n}_{3,s_4}$

$\dot{n}_{,w_1} = 4\dot{n}_2 + \dot{n}_{3,w_4} \rightarrow \boxed{.965 = 4\dot{n}_2 + \dot{n}_{3,w_4}}$ ✓

$\dot{n}_1 = 4 n_2 + \dot{n}_{3,10} \rightarrow \boxed{1 = 4 n_2 + \dot{n}_{3,4}}$

where is yield equation? -1

1) 22/25

2) 23/25

3) 18/25

4) 5/25

c) $\dot{m}_{10} = \frac{.035 \text{ wt\%}}{.05} = .7 \text{ Kg/s}$ in the final evaporator

69/100

$.965 = 10\dot{n}_2 + .95(.7)$

$.3 = 10\dot{n}_2$

$n_2 = .03 \text{ Kg/s}$ so we recover .3 Kg per 1 Kg

$\frac{.3 \text{ Kg}}{1 \text{ Kg}} = \boxed{.3}$ → This should be water content

So = $\underbrace{1 \times 0.965}$ → (1-0.035)

-2

$\dot{m}_4 = .88 \text{ Kg/s} = 1 - .12$

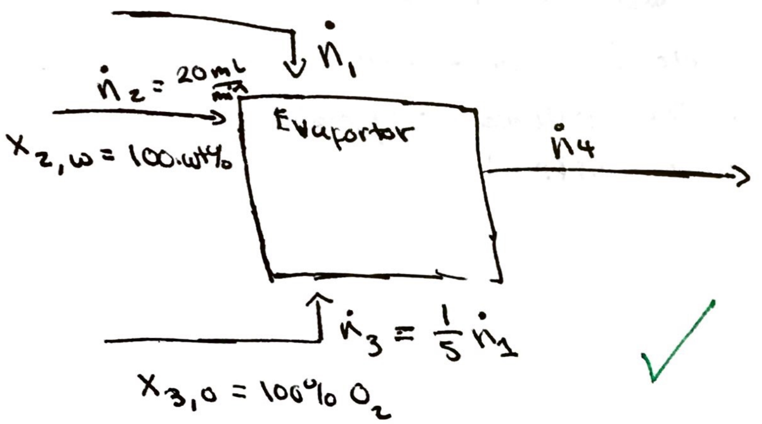
$.035 = (.88)\, \dot{m}_{3,4}$

$\dot{n}_{3,4} = .03977$ or $\boxed{3.977 \text{ wt\% salt}}$

You were supposed to solve the equations for 2 specified values!

2)

a) $X_{1,O} = 21 \text{ mol\%}$ $\quad X_{1,N_2} = 79\% \, N_2$ (mol %)



$X_{4,O} = ?$

$X_{4,w} = 1.5 \text{ mol\% water}$

$X_{4,N} = ?$

b) $n_{df} = n_{unknowns} - n_{indep\ eqns}$

$= 4 - (2-1-1) = 0$

$6 - 6 = 0 \quad -2$

23/25

c)

	$\dot{n}_1$	$\dot{n}_2$	$\dot{n}_3$	$\dot{n}_4$
O:	$.21\dot{n}_1$	$0\dot{n}_2$	$\dot{n}_3 \ (\frac{1}{5}\dot{n}_1)$	$X_{4,O}$
N:	$.79\dot{n}_1$	$0\dot{n}_2$	$0\dot{n}_3$	$X_{4,N}$
w:	$0\dot{n}_1$	$\dot{n}_2$	$0\dot{n}_3$	$.015\dot{n}_4$

~~$20 \frac{mL}{min} = .015\dot{n}_4 = \dot{n}_4 = 1333.33 \frac{cm^3}{min}$~~

$20 \frac{g}{min} \left(\frac{1 \text{ mol}}{18.02 \text{ g}}\right) = 1.111 \text{ mol/mins}$

$1.111 \text{ mol/min} = .015\dot{n}_4$

$\dot{n}_4 = 74.07 \text{ mol/min}$

$.21\dot{n}_1 + \frac{1}{5}\dot{n}_1 = X_{4O}\dot{n}_4 \rightarrow .41\dot{n}_1 = X_{4O}\dot{n}_4$

$.79\dot{n}_1 = X_{4N}\dot{n}_4 \rightarrow .79\dot{n}_1 = \dot{n}_4(.985 - X_{4O}) \rightarrow .79\dot{n}_1 = .985\dot{n}_4 - X_{4O}\dot{n}_4$

$-.79\dot{n}_1 + .985\dot{n}_4 = X_{4O}\dot{n}_4$

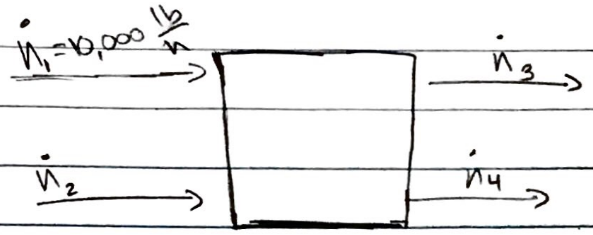
$.41\dot{n}_1 = -.79\dot{n}_1 + .985\dot{n}_4$

$1.2\dot{n}_1 = .985(74.07)$

$\boxed{\begin{array}{ll} \dot{n}_1 = 60.799 \text{ mol/min} & X_{4O} = .3365 \\ \dot{n}_3 = 12.159 \text{ mol/min} & X_{4N_2} = .6484 \end{array}}$

Sai Dereddy

8 10,000 lb/h   40 wt% TiO_2, 20 wt% salt, 40 wt% ~~water~~



$\dot{n}_1$ is the raw pigment

$x_{1,TiO_2} = 40$ wt%

$x_{1,salt} = 20$ wt%

$x_{1,water} = 40$ wt%

$\dot{n}_1 = 10{,}000\ \frac{lb}{hr}$

$\dot{n}_2$ is the water feed

$\dot{n}_2 = 6\dot{n}_1$

$x_{2,w} = 100$ wt%

$\dot{n}_3$ is the TiO_2 and salt

$x_{3,TiO_2} = 50$ wt% TiO_2

$x_{3,salt} = < 100/1000000$ salt

$x_{3,water} \approx 50$ wt% water

$\dot{n}_4$ = Water with salt

$x_{4,w} = 99.5\%$

$x_{4,salt} = < .5\%$

Dof =

4a - 9	5a - 2
4b - 0	5b - 3
	5c - 4
	5d - 3
9	12

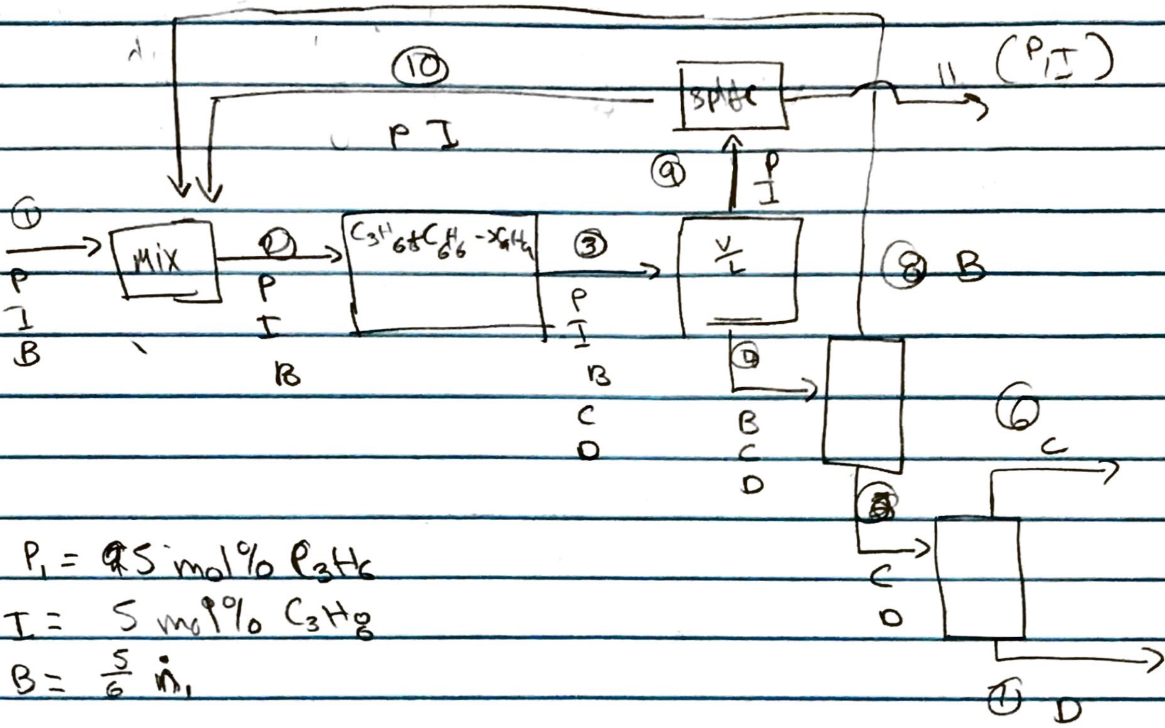
18/25

9 - 12 = -3

One way to fix the Dof is to get rid of the constraints. By getting rid of the ~~co~~ constraint we would know ~~know~~ much salt would be in each stream. Allowing us to solve for them

How are you going to get rid of constraints? Or which constraints are you going to get rid of? How do you decide which constraint is more important?

4)



$P_1 = 95$ mol% C_3H_6

$I = 5$ mol% C_3H_8

$B = \frac{5}{6}\dot{n}_1$

$I_2 = 10\%$ C_3H_8

80% of P and 72% of B are converted

Dof

4A : 25

4B : 2

Sa : 1

Sb : 4

Sc : 2

Sd : 20

$27 - 27 = 0$ ✓ specified

equations?

solving?

5/25